

FORM TP 2011180



TEST CODE **02220020**

MAY/JUNE 2011

**CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION**

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

20 MAY 2011 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do not open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

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MODULE 1

Answer **BOTH** questions.

1. (a) List **THREE** characteristics of commercial agricultural systems. [3 marks]
- (b) (i) Give **TWO** examples of the use of technology in agriculture. [2 marks]
- (ii) For **EACH** example given in (i) above discuss how it can be used to improve agricultural productivity. [4 marks]
- (c) Figure 1 shows the agricultural yield obtained by a farmer when using inorganic fertilisers and organic fertilisers on his farm.

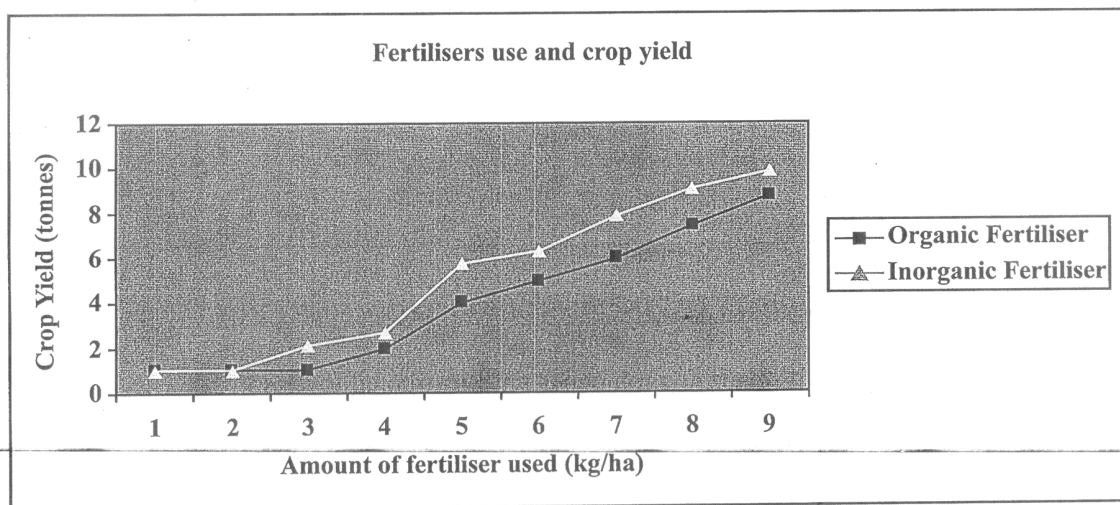


Figure 1. Agricultural yield obtained by a farmer when using inorganic fertilisers and organic fertilisers

- (i) Based on Figure 1 make **FIVE** deductions about the agricultural yield from the farm. [5 marks]
- (ii) Give **THREE** reasons why organic fertiliser is promoted as a feasible alternative to inorganic fertiliser despite the trend in Figure 1. [6 marks]

Total 20 marks

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2. Figure 2 shows the contributions made by subsistence and commercial agriculture to agro-processing, food security, GDP and employment in a Caribbean country.

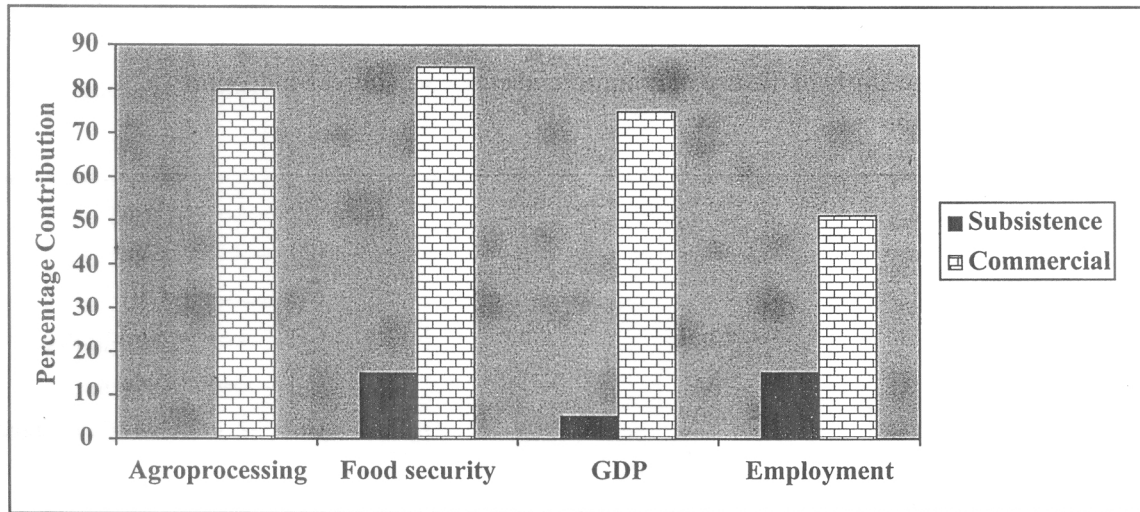


Figure 2. Contributions made by subsistence and commercial agriculture

- (a) (i) Based on Figure 2 make FOUR deductions about the contribution of subsistence and commercial agriculture to agro-processing, food security, GDP and employment. **[4 marks]**
- (ii) What is the total percentage contribution made by subsistence agriculture in the Caribbean country? **[1 mark]**
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- (b) State THREE features of sustainable agriculture. **[3 marks]**
- (c) Give FOUR reasons why climate change is a major threat to sustainable agriculture in the Caribbean. **[8 marks]**
- (d) Give TWO reasons why agro-forestry is accepted as an environmentally sustainable practice in agricultural systems. **[4 marks]**

Total 20 marks

MODULE 2

Answer BOTH questions.

3. Figure 3 shows the shifts in the use of commercial energy resources in Country Y since 1800, with projections to 2100.

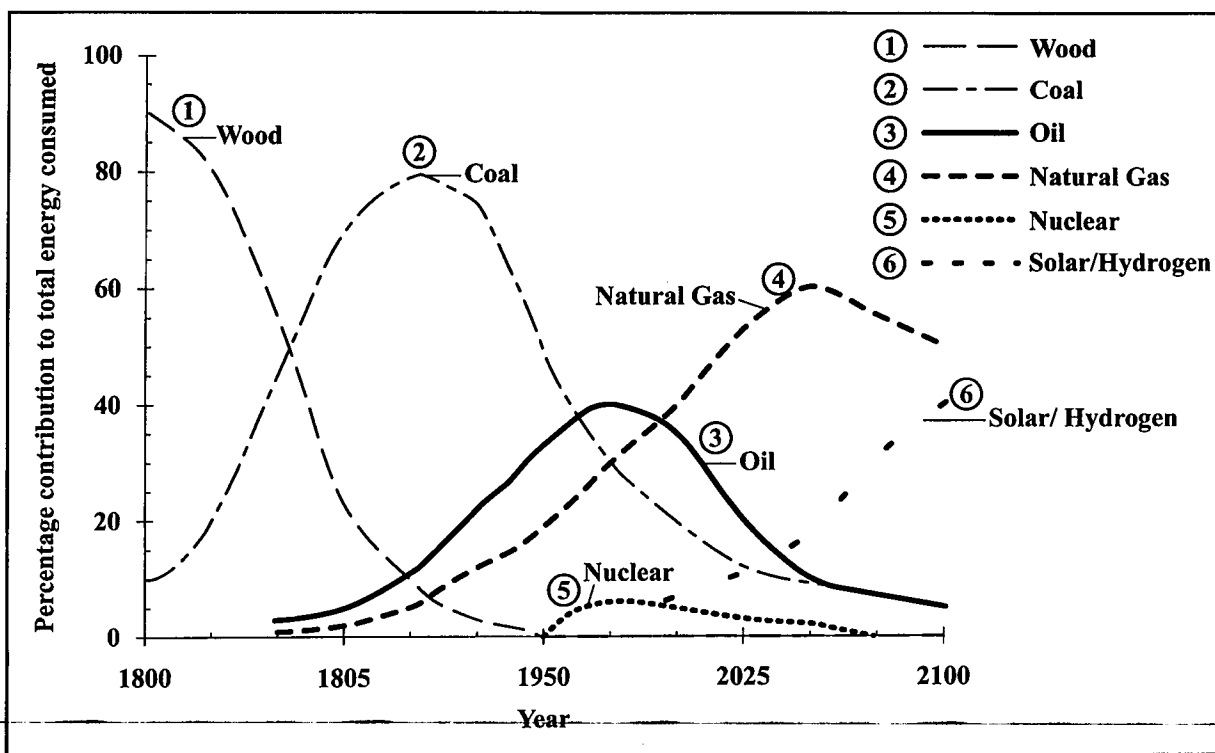


Figure 3. Shifts in the use of commercial energy resources in Country Y

- (a) Describe the pattern of commercial energy resource use in Country Y. [2 marks]
- (b) The energy resources illustrated in Figure 3 can be classified as renewable and non-renewable.
- (i) Which of the commercial energy resources illustrated in Figure 3 is expected to experience increased usage in the future? [1 mark]
- (ii) Into which classification does the energy resource described in (b) (i) above fall? [1 mark]
- (iii) Using the information in Figure 3, determine which commercial energy resource(s) may be in complete disuse by 2100. [2 marks]
- (iv) Give an explanation for the answer you provided in (b) (iii). [4 marks]

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- (c) For many Caribbean countries energy use is very costly.
- (i) Which of the resources illustrated in Figure 3 would you recommend for increased investment in the Caribbean? Give ONE reason for your answer. **[4 marks]**
 - (ii) State TWO limiting factors for the use of the energy resource recommended in (c) (i) above in a **named** Caribbean country. **[2 marks]**
- (d) Using alternative energy sources is one method of improving energy conservation and efficiency. **Other than the source mentioned in (c) (i) above**, describe ONE method for improving energy conservation, including a discussion of how relevant this method is for Caribbean countries. **[4 marks]**

Total 20 marks

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4. (a) (i) Use a flow diagram to illustrate the conventional generation of electricity using natural gas as the fuel. [4 marks]
- (ii) Annotate the diagram to indicate the function of EACH unit in the process. [3 marks]
- (b) Table 1 below provides data on the generation and consumption of electricity in various Caribbean countries in 2006.

**TABLE 1: ELECTRICITY GENERATION AND CONSUMPTION
IN VARIOUS CARIBBEAN COUNTRIES, 2006**

Country	Electricity Generation, billion kWh	Electricity Consumption, billion kWh
Antigua and Barbuda	0.11	0.10
Barbados	0.98	0.91
Belize	0.21	0.19
Dominica	0.09	0.08
Grenada	0.17	0.14
Guyana	0.90	0.75
Jamaica	7.04	6.10
St. Kitts and Nevis	0.13	0.12
Trinidad and Tobago	6.62	6.23

- (i) Classify the countries above in terms of the magnitude of their electricity generation, using the criteria in Table 2 on page 7. Present your answer in a suitable table.

TABLE 2

Size	Electricity generation range, billion kWh
Small	< 0.80
Medium	0.80 – 5.00
Large	> 5.00

[3 marks]

- (ii) Select ONE country from each of the SMALL and LARGE size classes. Suggest an explanation for the difference in electricity consumption between the two countries. [4 marks]

Figure 4 below shows the electricity generation and consumption data for three Caribbean countries.

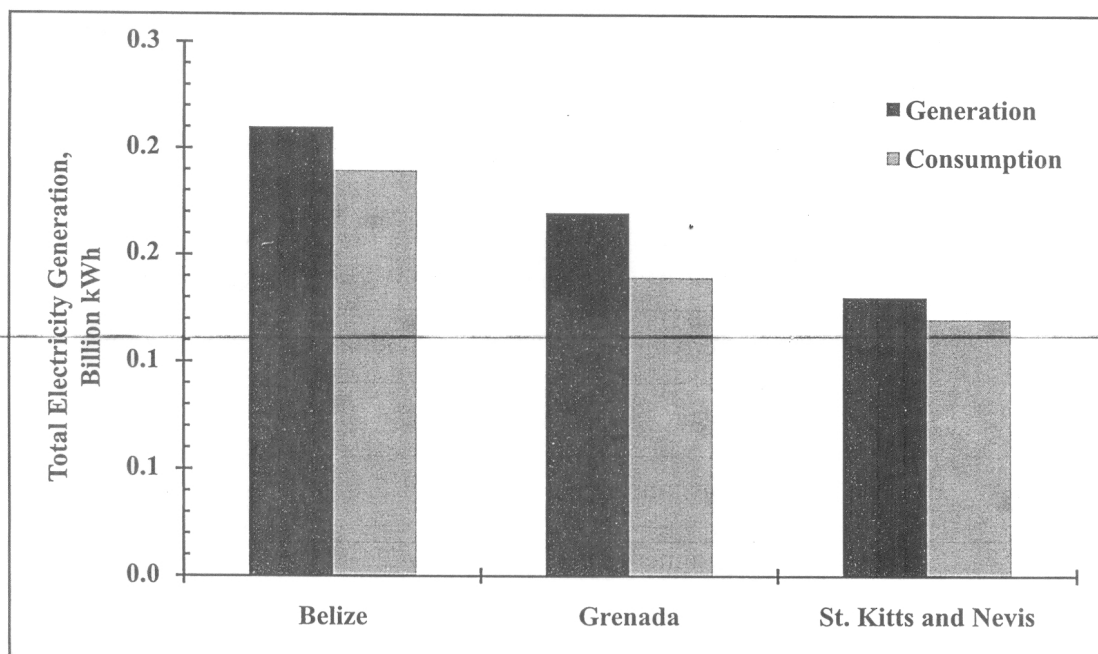


Figure 4. Electricity generation and consumption data for three Caribbean countries in 2006

- (c) What is the difference in electricity generation between Grenada and St. Kitts and Nevis? [3 marks]
- (d) In the Caribbean, many countries utilise fossil fuels as the major energy source in the electricity generation process. Outline ONE impact of fossil fuel use on the environment. [3 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions.

5. (a) What are 'secondary air pollutants'? [3 marks]
- (b) Figure 5 is an insert which shows an incomplete pathway of a pesticide when sprayed aerially. Complete Figure 5 to show ONE pathway by which the pesticide may enter
- (i) humans
- (ii) oceans. [2 marks]
- (c) Identify the processes in Figure 5 which are labelled A, B, C and D. [4 marks]
- (d) Table 3 presents results from a study that was conducted to determine the levels of a pesticide in a canal that is adjacent to an agricultural farm. The concentration of the pesticide in the water was determined to be 0.0003 ppm. During the study, aquatic organisms were sampled and the trophic structure of the aquatic ecosystem was constructed. The concentration of the pesticide in the organisms at EACH trophic level was measured.

TABLE 3: CONCENTRATION OF PESTICIDE IN ORGANISMS FROM AQUATIC ECOSYSTEM

Trophic Level	Pesticide concentration (ppm)
Producers	0.06
Primary consumers	0.4 – 1.3
Secondary consumers	1.5 – 2.7
Tertiary consumers	4.2 – 6.1

- (i) State THREE inferences that may be drawn from the results presented in Table 3. [3 marks]
- (ii) Calculate the minimum concentration factor of the pesticide in the tertiary consumers. [2 marks]
- (iii) Explain the pattern of pesticide concentration in the ecosystem that is shown in Table 3. [6 marks]

Total 20 marks

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Insert for Question 5. (b) and 5. (c)

Title:

Centre Number

Candidate Number

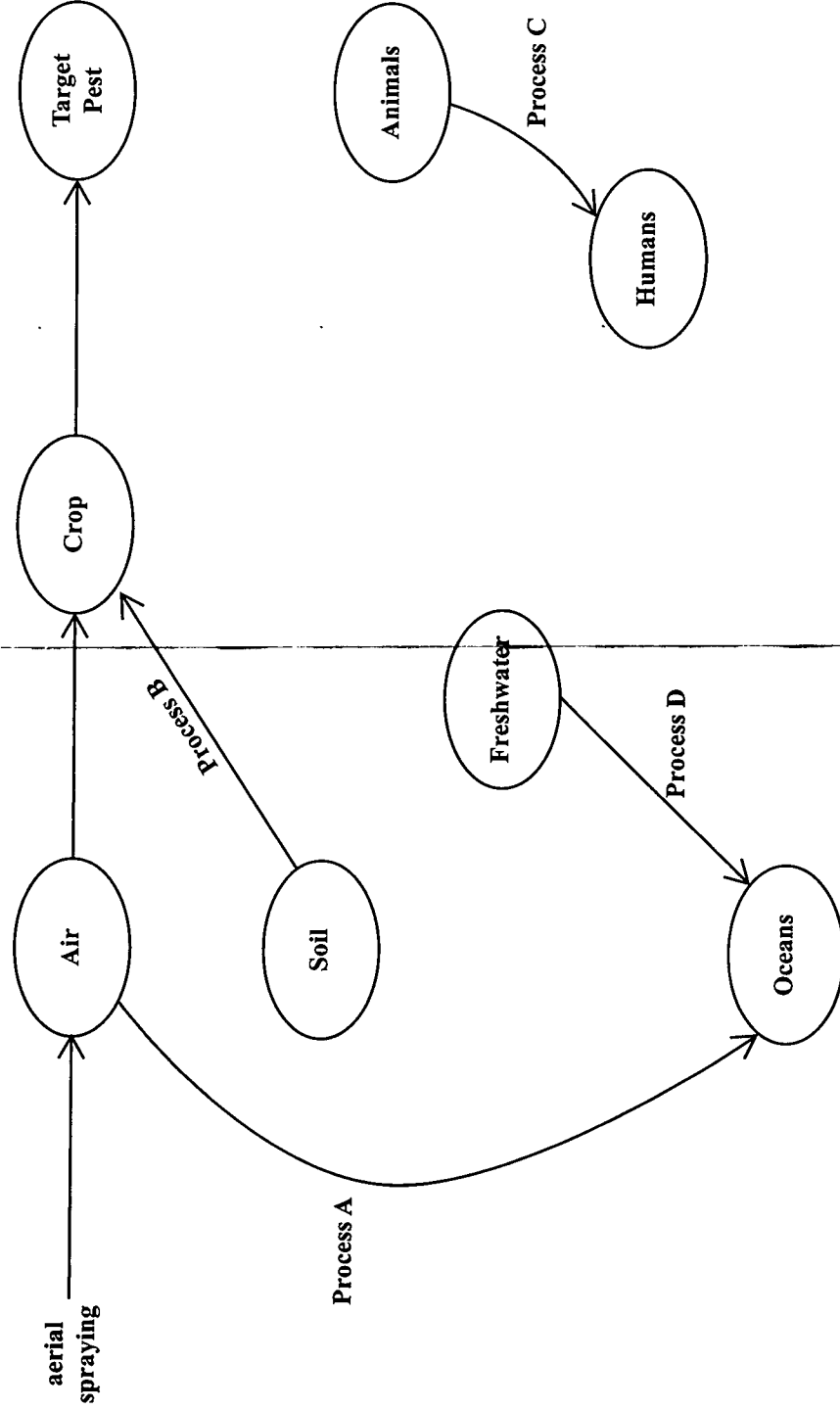


Figure 5. Pathway of pesticide after aerial spraying

6. (a) Figure 6 shows the amount of waste produced by three countries for the period 1965 to 2005. Study the diagram and answer the questions that follow.

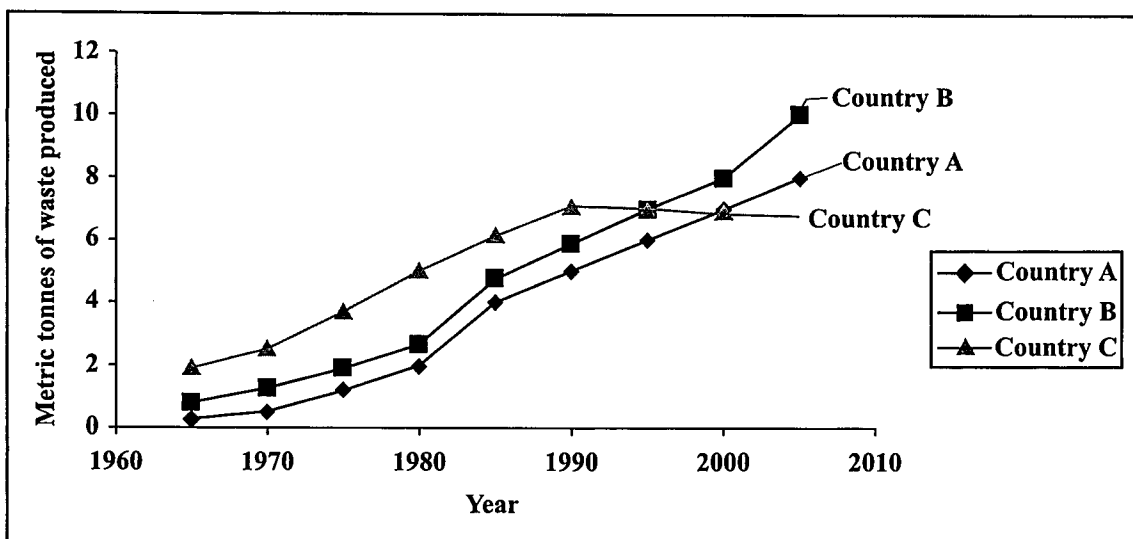


Figure 6. Amount of waste produced by three countries for the period 1965 to 2005

- (i) List THREE categories of waste produced in the Caribbean. [3 marks]
- (ii) Make THREE deductions from the information presented in Figure 6. [3 marks]
- (iii) Based on the information in Figure 6 estimate the projected waste production in 2010 for Country A. [2 marks]
- (b) Assess the effectiveness of EACH of the following as mitigation measures or solutions for environmental pollution.
- (i) Legislation [4 marks]
- (ii) Policy incentives [4 marks]
- (iii) Public awareness and education [4 marks]

Total 20 marks

END OF TEST